Logic in Computer Science, Prolog Test 1

BITS Pilani, K K Birla Goa Campus

Total: 10 Marks — Time : 45 Minutes

Course No: *CS F214* — September 25, 2025

Note:

- **Store your file using your ID number. For example if your ID no is 2014A7PS1234G then your file name should be 2014A7PS1234G.pl**
- **There should be no compilation error or warning messages when you run your prolog program. Any error or warning message, your program will not be evaluated.**
- **Your file should contain predicates (mydescendant/2, mymember/3, mydelete/3, mynextto/3, myhp/2, mynhp/2).**
- **You have to provide definitions of all the auxiliary predicates used and you are not allowed to use any libraries.**
- **The test cases might have variables as well not only just ground queries.**

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1. Write a prolog program with the following predicates
   (a) mydescendant(X,Y) returns true iff X is a descendant of Y. You can use the predicate parent(U,V) (here U is a parent of V and don't add any extra facts). We are assuming no one is descendant of himself/herself.

   (b) mymember(X,L,N) returns true iff X occurs in the list L at least N times. For example, mymember(a,[a,b,a,1],2) returns true and mymember(d,[d,a,b,1],2). While evaluating this question, the second and third parameter of mymember in the test case will not have variables. returns false.

   (c) mydelete(A,L1,L2) returns true iff A is an element of L1 and L2 the result of deleting the first occurrence of A from L1. For example, mydelete(4,[2,4,3,4,2],[2,3,4,2]) returns true,
   mydelete(4,[2,4,3,4,2],[3,2,4,2]) returns false,
   mydelete(4,[2,4,3,4,2],[2,3,2]) returns false,
   mydelete(4,[2,4,3,4,2],[2,4,3,2]) returns false and
   mydelete(5,[2,4,3,4,2],[2,4,3,4,2]) returns false.

   (d) mynextto(L,A,B) returns true iff A and B are adjacent in the list L. For example mynextto([1,2,3],2,1) returns true and mynextto([1,2,3],1,3) returns false. While evaluating this question, the first parameter of mynextto in the test case will not have variable.

   (e) Lists of decimal digits is used to represent positive numbers in this question. myhp(N1,N2) returns true iff N1 is permutation of N2 and the positive number represented by N1 is greater than the positive number represented by N2. For example myhp([2,3,1],[2,1,3]) returns true and myhp([2,0,4,1],[2,1,4,0]) returns false. While evaluating this question, the first parameter of myhp in the test case will not have variable.

   (f) **(Bonus question:Next Higher Permutation)** mynhp(N1,N2) returns true iff myhp(N1,N2) is true and for every other N, if myhp(N,N2) then the positive number represented by N is greater than the positive number represented by N1.